

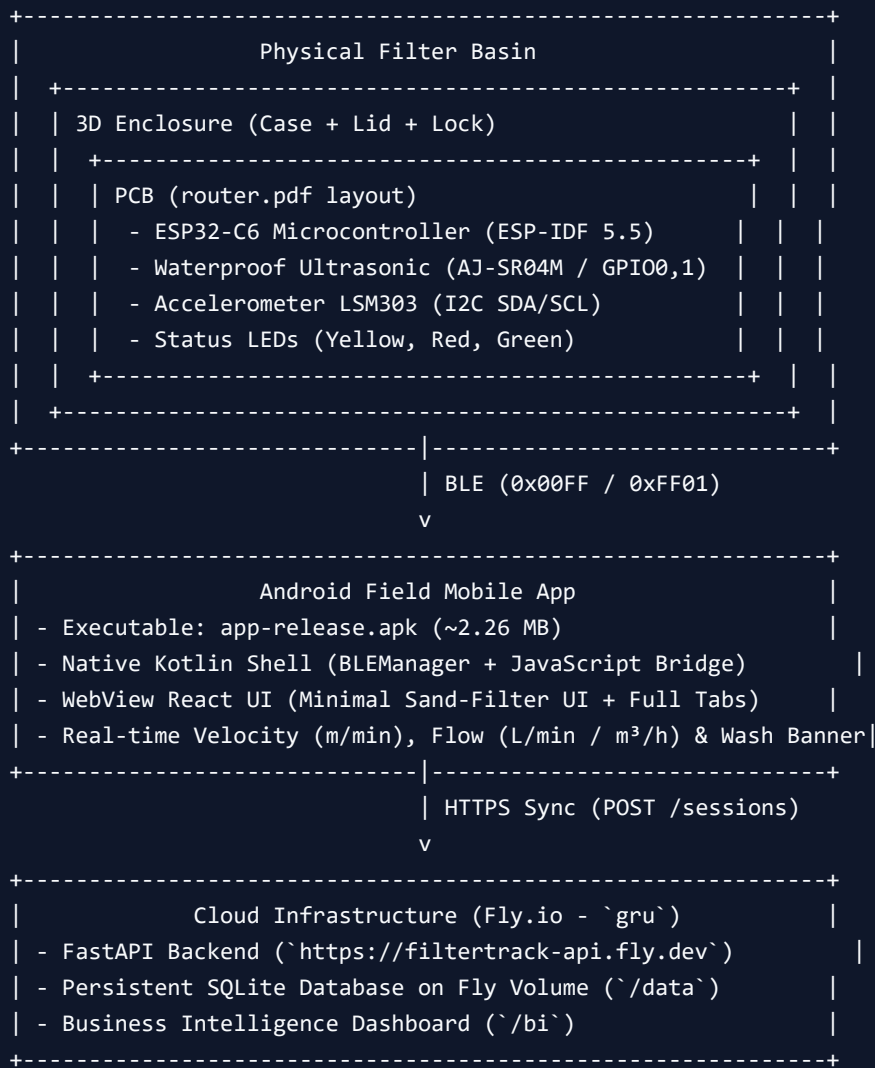
FilterTrack - Master Project Documentation

<i>Single Source of Truth</i> for the FilterTrack Water Flow Monitoring System
Primary Microcontroller: ESP32-C6
Document Status: Complete & Authoritative

1. Executive Summary & System Overview

FilterTrack is an end-to-end industrial IoT solution engineered for real-time water level and flow monitoring during sand-filter backwash operations at water treatment facilities (such as **Cagece**).

The system measures the rate of rise and fall of water levels inside filtration basins using an ultrasonic sensor backed by accelerometer-based tilt correction. Technicians capture field measurements using a custom Android mobile application connected via Bluetooth Low Energy (BLE). The mobile app processes real-time sensor streams, computes flow metrics, manages local measurement sessions, and syncs data to a cloud-hosted FastAPI backend with an integrated Business Intelligence (BI) dashboard.



2. Hardware Architecture & Main Microcontroller (ESP32-C6)

The definitive primary microcontroller for FilterTrack is the **Espressif ESP32-C6** (32-bit RISC-V single-core CPU up to 160 MHz, 4 MB Flash, supporting 2.4 GHz Wi-Fi 6, BLE 5.0, and 802.15.4 Thread/Zigbee).

2.1 Hardware Connections & Pinout

The **ESP32-C6** carrier board interfaces with the ultrasonic transducer, tilt accelerometer, status LEDs, and power supply according to the following GPIO configuration:

Peripheral / Component	Function	ESP32-C6 GPIO	Technical Description
Ultrasonic Sensor (AJ-SR04M)	Trigger	GPIO1	Output pulse (~10µs) to initiate measurement
Ultrasonic Sensor (AJ-SR04M)	Echo	GPIO0	Input pulse width proportional to target distance
Status LED - Yellow	Low-Power	GPIO2	High when ESP32-C6 is in low-power standby mode
Status LED - Red	Error / Adv	GPIO3	Blinks when advertising/disconnected; Solid on error
Status LED - Green	BLE Connected	GPIO4	Solid high when BLE GATT client (Android) is connected
Accelerometer (LSM303DLHC)	I2C SDA	GPIO13	I2C Data line for 3-axis accelerometer / magnetometer
Accelerometer (LSM303DLHC)	I2C SCL	GPIO12	I2C Clock line for 3-axis accelerometer / magnetometer
Sensor Power Control	Power Enable	<i>Disabled</i>	SENSOR_PWR_ENABLED 0 (direct continuous rail)

WARNING:

ESP32-C6 Native USB Pin Conflict Notice (GPIO12 / GPIO13):

On the ESP32-C6 silicon, **GPIO12 (D-)** and **GPIO13 (D+)** are hardwired to the native USB-Serial/JTAG controller. Using these pins for the accelerometer I2C bus (SDA/SCL) creates a physical bus conflict whenever a USB cable or USB host is connected, leading to `RAW=ERRO` payload responses.

Recommended Fix for Future PCB Revisions: Reassign I2C SDA/SCL to un-multiplexed GPIO pins such as **GPIO6** and **GPIO7**.

2.2 ESP32-C6 Flash Partition Layout (`partitions_ota.csv`)

The ESP32-C6 utilizes a 4 MB flash layout optimized for dual Over-The-Air (OTA) firmware updates and on-chip raw sensor logging:

Partition Label	Subtype	Flash Offset	Size	Purpose
nvs	nvs	0x009000	16 KB	Non-Volatile Storage (sand-mode state, calibration, logs)
otadata	ota	0x00D000	8 KB	OTA selection state & bootloader pointers
phy_init	phy	0x00F000	4 KB	PHY initialization data
ota_0	ota_0	0x010000	1344 KB	Primary Application Slot A (~28% headroom)
ota_1	ota_1	0x160000	1344 KB	Secondary Application Slot B (OTA target slot)
storage	undefined	0x2B0000	1344 KB	Raw Sensor Log Partition (Direct partition read/write/erase)

2.3 Firmware Compilation & Flashing Instructions

Build Environment: ESP-IDF v5.5.1 with `riscv32-esp-elf` cross-toolchain on Windows.

1. Building for ESP32-C6:

```
`powershell
```

```
# Navigate to the firmware directory
```

```
Set-Location C:\Users\Isaque\AndroidStudioProjects\FilterTrack\firmware
```

```
# Execute ESP-IDF build targeting ESP32-C6
```

```
& "C:\Espressif\python_env\idf5.5_py3.14_env\Scripts\python.exe" `
```

```
"C:\Espressif\frameworks\esp-idf-v5.5.1\tools\idf.py" `
```

```
-B build-esp32c6 -DSDKCONFIG=sdkconfig_esp32c6 build
```

```
`
```

Output binary: `firmware/build-esp32c6/FilterTrackv3.bin`

2. Flashing via Serial Cable (`esptool.py`):

```
`powershell
```

```
python -m esptool --chip esp32c6 -p COM3 -b 460800 `
```

```
--before default-reset --after hard-reset write-flash `
```

```
--flash-mode dio --flash-size 4MB --flash-freq 80m `
```

```
0x0 build-esp32c6\bootloader\bootloader.bin `
```

```
0x8000 build-esp32c6\partition_table\partition-table.bin `
0xd000 build-esp32c6\ota_data_initial.bin `
0x10000 build-esp32c6\FilterTrackv3.bin
`
```

3. Printed Circuit Board (PCB) Routing & Physical Design (`router.pdf`)

The printed circuit board for the FilterTrack ESP32-C6 sensor unit was custom designed and routed by Isaque (composite routing diagram documented in `router.pdf`).

3.1 Composite Drawing Details (`router.pdf`)

* **Document Name:** `router.pdf`

* **Title:** Composite Drawing (Printed Circuit Board Copper Routing)

* **Author:** Isaque (Created 2026-07-07)

* **Content & Layout:**

- Standard dual-layer PCB copper track layout connecting the ESP32-C6 microcontroller dev module headers to external peripherals.
 - Direct trace routing for **AJ-SR04M / JSN-SR04T** waterproof ultrasonic module connector (`GPI01` Trigger, `GPI00` Echo).
 - Dedicated header connections for **LSM303DLHC** 6-DOF accelerometer module (`GPI013` SDA, `GPI012` SCL).
 - Integrated current-limiting resistors and pads for status indicator LEDs (`GPI02` Yellow, `GPI03` Red, `GPI04` Green).
 - On-board power management traces delivering regulated +5V and +3.3V DC to sensor components with decoupling capacitors for ultrasonic noise reduction.
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4. 3D-Printed Protective Enclosure (.STL Files)

FilterTrack includes a complete 3-piece weather-resistant 3D-printed enclosure designed to protect the ESP32-C6 PCB, sensors, and power source when deployed over water filter basins.

STL File Name	File Size	Triangle Count	Approximate Dimensions (L × W × H)	Mechanical Function
FilterTrackCase.stl	91.9 KB	1,838	100 mm × 110 mm × 50 mm	Main Body Housing: Encloses the ESP32-C6 carrier PCB, battery compartment, status LED viewports, and ultrasonic sensor port.
FilterTrackLid.stl	27.0 KB	540	100 mm × 110 mm × 6 mm	Top Protection Cover: Snaps/screws onto the main case to seal against water splashes and debris.
FilterTrackLock.stl	948.7 KB	18,974	100 mm × 90 mm × 90 mm	Mounting Latch / Clamp: High-precision heavy-duty locking bracket for securing the unit onto filter basin railings.

4.1 3D Printing Guidelines

- * **Recommended Material:** PETG or ABS (for outdoor UV resistance, splash resistance, and structural durability).
- * **Layer Height:** 0.20 mm (Standard resolution).
- * **Infill:** 20% to 30% (Grid or Gyroid infill for structural strength on the mounting clamp).
- * **Wall Perimeter Count:** Minimum 3 to 4 walls (1.2 mm - 1.6 mm thickness).

5. Firmware BLE Contract & Communication Protocol

5.1 BLE Service & Characteristics

- * **Advertised Device Name:** FilterTrackV3
- * **Primary Service UUID:** 000000ff-0000-1000-8000-00805f9b34fb (0x00FF)
- * **Command & Notification Characteristic:** 0000ff01-0000-1000-8000-00805f9b34fb (0xFF01)
- * **OTA Data Characteristic:** 0000ff02-0000-1000-8000-00805f9b34fb (0xFF02)

5.2 Notification Data Formats (~100 ms Interval)

1. Standard Reading (with raw tilt accelerometer):

`\text`

`DIST=%.2f;ACC_RAW=%d,%d,%d`

```

**Compact split chunk format:** `D=35.47;A=12,-8,102` (reassembled in `BLEManager.kt` upon receiving `D=` ).

2. Tilt Distance Correction Formula:

When tilt accelerometer data is available, Android computes true vertical distance using:

$$\text{Vertical Distance} = \text{DIST} \times \sin\left(\arccos(|g_z|)\right)$$

5.3 Control Commands (Android to ESP32-C6 over `0xFF01` )

| Command String                  | Firmware Response / Action                                                 | Application Usage                                                |
|---------------------------------|----------------------------------------------------------------------------|------------------------------------------------------------------|
| <code>0</code>                  | Calls <code>esp_restart()</code>                                           | Reboots the ESP32-C6 microcontroller                             |
| <code>1</code>                  | Enables low-power standby; turns off ultrasonic sensor                     | Saves power between operational shifts                           |
| <code>2</code>                  | Disables low-power standby; powers ultrasonic sensor back on               | Resumes active measurement                                       |
| <code>3</code>                  | Notifies <code>VER=&lt;PROJECT_VER&gt;</code>                              | Query current firmware version                                   |
| <code>4</code>                  | Runs full flash self-test on <code>storage</code> partition                | Hardware validation (destroys logged raw data)                   |
| <code>5</code>                  | Enables sand-filter wash detection mode ( <code>SAND=1</code> )            | Enforces automatic wash detection state machine                  |
| <code>6</code>                  | Disables sand-filter mode ( <code>SAND=0</code> )                          | Aborts open wash state without logging records                   |
| <code>LOG:COUNT</code>          | Notifies <code>WL=CNT,&lt;count&gt;,&lt;boot&gt;,&lt;uptime&gt;</code>     | Queries stored wash log entry count                              |
| <code>LOG:READ</code>           | Streams <code>WL=R,...</code> records followed by <code>WL=END</code>      | Dumps logged wash events to Android app                          |
| <code>LOG:ACK:&lt;n&gt;</code>  | Erases log if count matches <code>&lt;n&gt;</code> ( <code>WL=CLR</code> ) | Confirms log sync and frees on-chip storage                      |
| <code>CAL:SET:&lt;cm&gt;</code> | Calibrates sensor tilt factor based on known water level                   | Calculates tilt factor $f = \frac{\text{real}}{\text{measured}}$ |

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## 6. Android Mobile Application & Release APK ( `app-release.apk` )

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The field technician mobile client is built as a hybrid native/web Android application.

### 6.1 Executable `.apk` and Package Information

- \* **Production Release APK File:** `app/build/outputs/apk/release/app-release.apk`
- \* **File Size: 2.26 MB (2,265,043 bytes)**
- \* **Build Target: Release Signed ( `versionCode=2` , `versionName=1.0` )**
- \* **Development Debug APK File:** `app/build/outputs/apk/debug/app-debug.apk` **(14.75 MB)**
- \* **Google Play Bundle:** `app/build/outputs/bundle/release/app-release.aab` ( `versionCode=3` , `versionName=1.1` )
- \* **Package Name:** `com.filtertrack`

### 6.2 Application Architecture & Native-JS Bridge

The app features a native Kotlin shell ( `MainActivity.kt` , `BLEManager.kt` , `BiDashboardActivity.kt` ) hosting an embedded WebView UI ( `app/src/main/assets/index.html` ).

- \* **JavaScript Bridge Name:** `window.Android`
- \* **Exposed Native Commands:** `startScan()` , `stopScan()` , `connect(address)` , `disconnect()` , `sendCommand(cmd)` , `openBiDashboard()` , `isBluetoothEnabled()` , `startFirmwareUpdate(bytes)` .
- \* **Native Callbacks to JS:** `window.FilterTrackBridge.onDeviceFound(...)` , `onDataReceived(...)` , `onConnectionStateChanged(...)` , `onOtaProgress(...)` .



## 6.3 Dual User Interface Modes

### 1. Minimal Mode (Default Field Interface):

- \* Optimized for quick field operation by plant technicians.
- \* Auto-sends command `5` ( `SAND=1` ) on connect to enable ESP32-C6 wash detection.
- \* Shows a clean single-screen interface:
  - **Velocidade de subida (Water Velocity):** Calculated in **m/min** (2 decimal places) over rolling 10-second analysis windows.
  - **Vazão (Water Flow Rate):** Displayed in **L/min** (or toggled to **m³/h**).
  - **Flow Direction Badge: Subindo (Green / Rising)** when level increases; **Descendo (Red / Falling)** when level decreases.
  - **Wash State Banner:** Listens for `WL=WSTART` and `WL=WEVT` notifications to announce active filter backwash sessions.
- \* Includes a button `"Abrir app completo"` to access advanced tools.

### 2. Full App Mode (Multi-Tab Navigation):

- \* **Monitor Tab:** Full real-time metric cards, velocity history chart, flow direction, and selected filter metadata.
- \* **Sessões Tab:** Local session archives, session detail chart modal, manual cloud sync triggering, and compact JSON exports.
- \* **Filtros Tab:** Filter catalog browser and filter selection picker with station/area details.
- \* **Ajustes Tab:** Firmware OTA update utility, sensor tilt calibration ( `CAL:` ), wash detection parameter configuration ( `CFG:` ), low-power control, and ESP restart button.
- \* **Dados Tab:** Launches the integrated Business Intelligence dashboard.

## 6.4 Data Filtering Rules & Signal Processing Pipeline

- \* **Minimum Distance Threshold:** Distance readings  $< 25\text{ cm}$  are discarded as invalid sensor noise.
  - \* **Outlier Filtering:** 2-second median/MAD windowing.
  - \* **Maximum Velocity Cap:** Velocities  $> 10\text{ m/min}$  are rejected as physical spikes.
  - \* **Movement Threshold:** Water level displacement  $< 0.1\text{ cm}$  over 10 seconds is treated as stationary ( $0.0\text{ m/min}$ ).
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## 7. Cloud Backend & BI Dashboard

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### 7.1 Backend Services & Hosting

- \* **Framework:** FastAPI (Python 3.11+)
- \* **Database:** SQLite on persistent Fly.io Volume ( `/data/filtertrack.db` )
- \* **Production API Base URL:** `https://filtertrack-api.fly.dev`
- \* **BI Dashboard Base URL:** `https://filtertrack-api.fly.dev/bi`

### 7.2 Session Upload Endpoint ( `POST /filtertrack/sessions` )

Field sessions are synced using a compact array serialization scheme ( `distance_cm_x100_v1` ) to minimize cellular payload sizes:

```

{
 "schemaVersion": 1,
 "app": "FilterTrack",
 "uploadedAt": "2026-08-07T10:40:00Z",
 "session": {
 "id": "session-1770000000000",
 "startedAt": "2026-08-07T10:30:00Z",
 "endedAt": "2026-08-07T10:40:00Z",
 "endReason": "manual_stop",
 "device": { "name": "FilterTrackV3", "address": "AA:BB:CC:DD:EE:FF" },
 "filter": {
 "id": "ETA_GAVIÃO_FILTRO_01",
 "name": "Filtro 01",
 "areaM2": 7.07,
 "station": "ETA Gavião",
 "businessUnit": "UN-METROPOLITANA"
 },
 "samples": {
 "format": "distance_cm_x100_v1",
 "t0": 1770000000000,
 "dtUnit": "ms",
 "distanceUnit": "cm",
 "scale": 100,
 "t": [0, 100, 200, 300],
 "d": [3547, 3546, 3544, 3542]
 }
 }
}

```

### 7.3 Integrated BI Dashboard ( `backend-fastapi/app/static/bi.html` )

The BI dashboard provides management and operational teams with comprehensive insights:

- \* **Key Performance Indicators (KPIs):** Total sessions, active filters monitored, average net backwash flow rates, total backwash volume.
  - \* **Interactive Charts:** Average Net Flow by Filter (L/min), Flow Direction Distribution, Daily Backwash Volume Trends, Samples by Water Treatment Station.
  - \* **Data Correction Workflows:** Allows operators to submit proposals for re-assigning filter metadata or splitting multi-phase sessions into discrete time slices. Admin role approves or rejects proposals.
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## 8. File & Directory Reference Catalog

| Directory / File                              | Description & Purpose                                                                                             |
|-----------------------------------------------|-------------------------------------------------------------------------------------------------------------------|
| DOCUMENTATION.md                              | <b>This master documentation file (Single Source of Truth)</b>                                                    |
| router.pdf                                    | Printed Circuit Board (PCB) composite copper routing diagram                                                      |
| FilterTrackCase.stl                           | Main body 3D enclosure STL file (100 × 110 × 50 mm)                                                               |
| FilterTrackLid.stl                            | Top cover 3D lid STL file (100 × 110 × 6 mm)                                                                      |
| FilterTrackLock.stl                           | Heavy-duty mounting latch/clamp STL file (100 × 90 × 90 mm)                                                       |
| app/build/outputs/apk/release/app-release.apk | <b>Production executable release APK</b> (~2.26 MB)                                                               |
| firmware/main/FilterTrackv3.c                 | Core ESP-IDF C source code for ESP32-C6                                                                           |
| firmware/partitions_ota.csv                   | ESP32-C6 flash partition table (OTA slots + sensor storage)                                                       |
| app/src/main/assets/index.html                | Native Android WebView user interface (Minimal + Full modes)                                                      |
| backend-fastapi/app/static/bi.html            | Static BI dashboard served at <a href="https://filtertrack-api.fly.dev/bi">https://filtertrack-api.fly.dev/bi</a> |

## 9. Conclusion & Verification

This master document represents the single, consolidated, true specification for the **FilterTrack** system as of August 2026. All hardware pinouts, software boundaries, physical enclosure files, and communication protocols defined herein reflect the active production system.